
Preserve, Align, Reject: Four Studies of Vision Models Beyond the IID Assumption

Mohid Ali

Lahore University of Management Sciences (LUMS)
mohidali669@gmail.com

Abstract

Standard supervised evaluation assumes that test inputs match the training distribution and that every input belongs to a class seen during training. This report studies four ways that assumption can fail, using a shared family of pretrained vision backbones. The first study compares a convolutional network, a Vision Transformer, and a contrastively trained CLIP model under controlled visual interventions to see which cues, shape, texture, color, and spatial layout, drive their predictions. The second study adapts a shared ResNet backbone to an unseen visual domain using unlabeled target examples, contrasting an explicit distribution alignment objective against two adversarial variants. The third study instead generalizes the same backbone to that domain with no target access at all, contrasting alignment across the labeled source domains against a sharpness based optimizer that never touches domain alignment. The fourth study evaluates a classifier trained only on known classes against novel categories, contrasting simple post hoc novelty scores against a method that alters training itself to make room for the unknown. Together, these studies raise three distinct questions: what a representation should preserve under a shift in domain, what it can safely become invariant to, and, in the fourth study’s open set setting, when it should decline to answer at all. Doing well on one gives little guarantee of doing well on the others. Code, configurations, and full result tables are available at https://github.com/Mohidali2005/ATML_PA1.

1 Introduction

Most machine learning benchmarks quietly assume that test data looks like training data and that every test example belongs to a class the model was trained to recognize. Neither assumption survives the real world: an object can appear in an unfamiliar color or pose, an entire visual domain can shift from photographs to sketches, and a system built around a fixed label set will eventually meet something it was never meant to recognize. This report studies four such departures from the closed set, identically distributed setting, using controlled experiments on a shared family of pretrained vision backbones.

Before asking how a model behaves once its input distribution changes, it is worth asking what it responds to when nothing has changed. A convolutional network such as ResNet [7], a Vision Transformer [2], and a contrastively trained model such as CLIP [16] rest on different inductive biases: local convolution, global self attention, and language supervised alignment respectively. Comparable clean accuracy is therefore no guarantee they rely on comparable evidence. A shape texture cue conflict methodology [5] gives a controlled way to ask this directly, and the first study extends it into a broader comparison of color, shape, and spatial sensitivity across all three architectures.

Knowing what a model relies on under one controlled intervention is a natural precursor to a harder question: what happens once an entire domain shifts? Unsupervised domain adaptation gives a model unlabeled access to the new domain, discouraging it from telling source and target apart through an explicit discrepancy measure [14] or an adversarial discriminator [4, 15]. Domain generalization asks a stricter version with no such access at all, hoping that training on multiple observed domains already transfers to an unseen one, a hope careful benchmarking suggests is not always well founded [6]. An alternative to hoping that variation across domains cancels out on its own is to seek parameters that are locally stable regardless of which domain they see [3]. The second and third studies place these two experiments side by side under an identical source domain protocol, isolating how much target domain access helps.

Domain shift still assumes every test image belongs to a known class; it only changes how that image looks. The more fundamental violation is an input belonging to no known class at all, something an ordinary classifier cannot refuse. Open set recognition studies this directly: a classifier’s closed set quality is itself informative about how well it rejects the unknown [19], while other approaches alter training to carve out room for rejection [20]. The fourth study compares both families on a shared classifier.

Together, these four studies raise three distinct questions: what a representation should preserve across a domain shift, what it can safely become invariant to, and, in the fourth study’s open set setting, when it should decline to force a prediction at all.

2 Shared Experimental Setup

The four studies share a common protocol: hold a backbone fixed across conditions, vary one factor at a time, and evaluate every condition on the same held out images with a fixed seed throughout. Task 1 uses three frozen, ImageNet pretrained backbones on a class balanced STL-10 subset [1]. Tasks 2 and 3 fine tune a single ImageNet pretrained ResNet-18 end to end on PACS’s seven classes [12], with Photo, Art Painting, and Cartoon as the labeled sources and Sketch as the evaluation target throughout. Both tasks reuse identical source splits and training budgets, so touching the target domain during training is the only real difference between them, and BatchNorm statistics stay frozen at their pretrained values throughout so BatchNorm cannot silently perform its own adaptation. Task 4 trains a CIFAR appropriate ResNet-18 from scratch on CIFAR-10 [10] and evaluates rejection of two fixed CIFAR-100 groups chosen for semantic closeness to, or distance from, the known categories. Each task’s own methodology below gives the class counts, splits, and hyperparameters in full.

3 Task 1: Inductive Biases and Feature Representations

3.1 Methodology

Three frozen pretrained backbones, ResNet, a Vision Transformer, and CLIP’s image encoder, each pair with a linear head trained on STL-10’s ten classes; CLIP is also evaluated zero shot with a fixed caption template. Every intervention is applied to a common image before backbone specific normalization. Grayscale and class swapped color transfer test color sensitivity, a coarse patch shuffle tests global layout, and reflected translation tests positional sensitivity. AdaIN style transfer [9] produces the fourth intervention, a cue conflict image pairing one class’s shape with another’s texture, following an established cue conflict protocol [5]. A structural similarity check between the content and stylized images in grayscale discards any pair where the original shape is no longer visible, and each surviving image is then scored as a shape response, a texture response, or neither:

$$\text{Shape bias} = \frac{N_{\text{shape}}}{N_{\text{shape}} + N_{\text{texture}}}, \quad \text{Coverage} = \frac{N_{\text{shape}} + N_{\text{texture}}}{N_{\text{total}}}. \quad (1)$$

Cosine similarity between clean and transformed features, and a joint t-SNE projection [18], visualize how each backbone’s representation changes under every intervention.

3.2 Results

Condition	ResNet	ViT	CLIP	Model	Shape bias (%)	Coverage (%)
Clean	0.986	0.982	0.978	ResNet head	68.6	75.5
Grayscale	0.968	0.956	0.968	ViT head	83.2	83.5
Color swap	0.986	0.978	0.980	CLIP head	84.9	87.6
Patch shuffle	0.910	0.922	0.880	CLIP zero shot	82.5	85.1

Table 1: Left: accuracy under each intervention; patch shuffle is the most damaging condition for every backbone, but not by the same margin. Right: shape bias and coverage from the cue conflict evaluation.

Table 1 shows grayscale costs a little accuracy, since removing color leaves each image’s shape and texture structure intact. Color swap costs almost nothing, since STL-10’s ten classes already separate well by shape and texture alone. Translation is similarly mild, since the object stays in frame at these displacement sizes with reflection padding keeping the background plausible. Patch shuffle hurts every backbone substantially, since it destroys global structure, and it hurts CLIP the most. All three backbones favor shape over texture on the cue conflict images, the opposite of the texture bias

reported for ordinary ImageNet trained CNNs [5]; this likely reflects retraining only a linear head here, on STL-10’s ten coarse superclasses instead of ImageNet’s thousand fine grained ones, which pushes a dog versus cat boundary toward shape like invariances almost by construction. Coverage is lowest for ResNet alongside its lowest shape bias, so its weaker shape preference looks like a property of the representation: a pure scoring coincidence would not track shape bias this closely across models. That aggregate coverage number still hides real variation at the level of single images: Table 2 shows one horse shaped, deer textured conflict read as deer by every model, while the most heavily stylized conflicts match neither label at all.

Shape	Texture	ResNet	ViT	CLIP head	CLIP zero shot
cat	dog	cat	cat	cat	cat
horse	deer	deer	deer	deer	deer
cat	dog	cat	dog	cat	dog
cat	dog	bird	bird	monkey	ship

Table 2: Four representative cue conflict predictions, from full shape agreement to a total breakdown where no model matches either label.

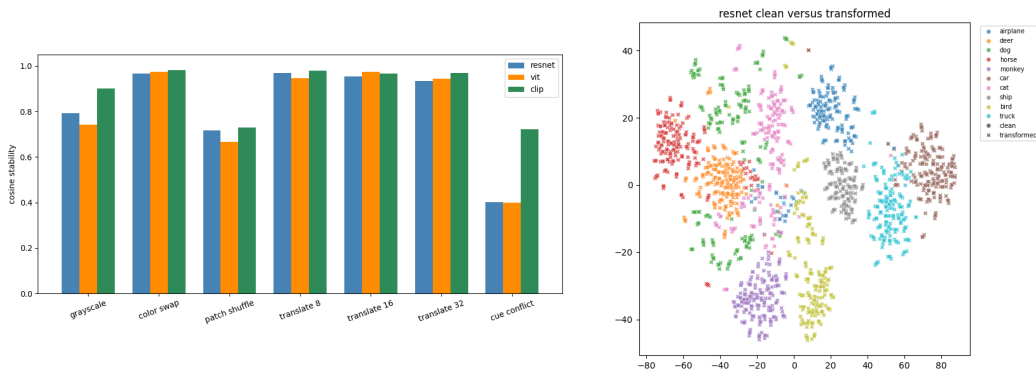


Figure 1: **CLIP’s representation is the most stable of the three backbones under every intervention, including the two where its own accuracy drops most.** Left: cosine similarity between each backbone’s clean and transformed features under every intervention. Right: a joint t-SNE projection of ResNet’s clean (circle) and transformed (cross) features, colored by class, for a pool combining grayscale, patch shuffled, translated, and cue conflict images.

Figure 1 shows prediction level and representation level evidence mostly agree: patch shuffle and cue conflict hurt accuracy most and also move features furthest from their clean counterparts. Transformed points in the t-SNE projection still cluster near their own class’s clean cluster far more often than they cross into another class’s region, which is consistent with prediction accuracy staying well above chance even under the largest feature shifts. CLIP is the exception to the stability pattern. Its features stay the most stable of the three under every intervention, which suggests the representation itself drives these outcomes more than the classifier built on top of it. Patch shuffling is also the one case that defies a simple locality story: CLIP suffers the largest accuracy drop despite the convolution free ViT having no locality prior at all. ViT’s positional embeddings plausibly help it reassociate shuffled patches, while CLIP, trained to match whole images against whole captions, depends more on the overall composition staying intact; that dependence points to CLIP’s contrastive objective as the likely source of its sensitivity to shuffled layout, a training signal ViT and ResNet never received [17].

4 Task 2: Unsupervised Domain Adaptation

4.1 Methodology

A single ImageNet pretrained ResNet-18 is fine tuned on PACS’s three labeled source domains; Sketch supplies unlabeled images during training and labeled images only at evaluation. Source-only cross entropy is the baseline against three alignment methods. DAN adds a multi kernel Maximum Mean Discrepancy penalty between source and target features before the classifier head [14]:

$$\mathcal{L}_{\text{DAN}} = \mathcal{L}_{\text{cls}} + \lambda_{\text{MMD}} \|\mathbb{E}_s[\phi(F(x_s))] - \mathbb{E}_t[\phi(F(x_t))]\|_{\mathcal{H}}^2. \quad (2)$$

MMD compares only a fixed, kernel based summary of the two feature distributions, so a domain difference that kernel cannot see goes unpenalized. DANN removes that dependence on a hand picked kernel by learning the comparison itself: a domain discriminator sits behind a gradient reversal layer, so backpropagation simultaneously minimizes classification loss and maximizes domain confusion under a schedule that ramps up reversal strength [4]. That added flexibility has a cost: it turns alignment into a minimax game between discriminator and feature extractor instead of a single smooth penalty, and minimax games are notoriously harder to stabilize. CDAN keeps the adversarial discriminator but targets a different weakness: matching the marginal feature distribution alone says nothing about whether classes line up across domains. It conditions the discriminator on the outer product of feature and predicted class distribution, $g(x) = \text{vec}(f \otimes p)$, so alignment follows class structure instead of the marginal distribution alone [15]. DANN and CDAN training also clips gradients to a norm of 5, L2 normalizes the features feeding the discriminator, and tunes the learning rate per method, $3\text{e-}5$ for DANN and the shared $1\text{e-}4$ for CDAN. A held out logistic regression probe distinguishing source from target features gives a domain separability score, where 50 percent is chance.

4.2 Results

Method	Mean source acc.	Target (Sketch) acc.	Domain separability	Target Δ
Source-only	0.957	0.616	0.996	–
DAN	0.942	0.643	0.894	+0.026
DANN	0.955	0.715	0.999	+0.099
CDAN	0.946	0.276	0.995	-0.340

Table 3: Task 2 method comparison. Only DAN’s separability drop tracks its own target accuracy gain; DANN and CDAN’s separability barely differs from source-only’s despite their adversarial objective, so their numbers cannot be read as evidence about alignment quality.

Source-only already reaches 0.616 target accuracy from cross entropy, well above chance, but leaves a near perfect domain gap (separability 0.996) that is also uneven across classes: guitar transfers almost perfectly since its outline barely changes across styles, while giraffe and dog lose most of their accuracy, plausibly leaning on color and texture cues Sketch lacks, a pattern invisible from source domain validation alone. DAN narrows the domain gap to 0.894 and lifts target accuracy to 0.643, but its per class numbers (Table 5) show the gain is not uniform: most classes improve, some substantially, while person collapses from 0.825 to 0.006, driven almost entirely by person sketches classified as house (Appendix Figure 4).

DANN and CDAN both reach mean source accuracy in line with the other two methods, 0.955 and 0.946, and DANN reaches 0.715 target accuracy, moving past source-only’s 0.616. CDAN’s target accuracy stays low at 0.276. Its confusion matrix (Appendix Figure 5) shows nearly every Sketch image predicted as dog regardless of true class, while its own source validation accuracy stays strong throughout. The failure is specific to the target domain, since the shared backbone itself is training normally.

The controlled sweep over DAN’s MMD weight reinforces the picture: at $\lambda_{\text{MMD}} = 0.1, 1.0$, and 10.0 , mean source accuracy is 0.9466, 0.9468, and 0.8834, target accuracy is 0.766, 0.661, and 0.654, and domain separability is 0.986, 0.882, and 0.875. Separability falls steadily as λ_{MMD} rises, but target accuracy is highest at the lowest weight tested, and mean source accuracy is statistically indistinguishable between $\lambda_{\text{MMD}} = 0.1$ and $\lambda_{\text{MMD}} = 1$ despite a ten point gap in target accuracy.

Lower domain separability tracks better target accuracy only for DAN; DANN and CDAN’s separability barely moves despite their adversarial objective, so their target accuracy differences track how well each classifier generalizes on its own instead of any working alignment mechanism, and even DAN’s own gain is uneven, since the same alignment that helps most classes drives person’s accuracy toward zero. CDAN’s class conditional discriminator is meant to prevent this kind of mixing, yet its target domain collapse onto a single class still happens, pointing to a degenerate solution specific to the unlabeled Sketch data that a class balanced source domain evaluation would never show.

5 Task 3: Domain Generalization

5.1 Methodology

Task 3 reuses Task 2’s exact source domain splits, backbone, and training setup, with one restriction: no Sketch image, labeled or not, reaches training, model selection, or hyperparameter choice; it

appears only in the final evaluation. Task 2’s source-only checkpoint carries over unchanged as the ERM baseline. DAN-DG applies the same MMD mechanism as DAN, but pairwise across the three *observed* source domains instead of between a source and the target,

$$\mathcal{L}_{\text{DAN-DG}} = \mathcal{L}_{\text{ERM}} + \frac{\lambda_{\text{DG}}}{3} \sum_{e < e'} \text{MMD}^2(F(X_e), F(X_{e'})), \quad (3)$$

testing whether invariance learned purely from the domains already available transfers to one never seen. Aligning features is not the only way to seek robustness: SAM instead asks whether the optimizer has settled somewhere flat or sharp on the loss surface, independent of how alike the domains look in feature space. It leaves the objective and the set of domains untouched and changes only how the optimizer searches, solving $\min_{\theta} \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}_{\text{ERM}}(\theta + \epsilon)$ [3] to seek parameters whose local loss surface stays flat across a neighborhood of the optimum. A local sharpness proxy, the increase in validation loss after one normalized gradient ascent step, is measured for all three models on a fixed held out batch.

5.2 Results

Method	Mean source acc.	Sketch acc.	Source separability	Sharpness
ERM	0.957	0.616	0.860	0.321
DAN-DG	0.948	0.707	0.727	0.334
SAM	0.967	0.679	0.820	0.110

Table 4: Task 3 method comparison. DAN-DG and SAM both beat the ERM baseline on Sketch through different mechanisms: DAN-DG lowers source separability, SAM lowers the sharpness proxy, and each method leaves the other’s diagnostic largely unchanged.

Table 4 shows both proposed methods beating ERM through visibly different mechanisms: DAN-DG lowers source domain separability substantially and delivers the larger Sketch accuracy gain of the two, while its sharpness proxy is no better than ERM’s own; SAM moves separability only slightly but drives sharpness down by nearly a factor of three, yielding a smaller but still real Sketch accuracy gain. Mean source accuracy is almost uninformative about this gap: all three methods sit within a point of each other, yet SAM, highest on source accuracy, is not highest on Sketch. Per class evidence (Table 5) shows DAN-DG’s gain concentrated on elephant and giraffe at the cost of a severe person collapse, strikingly similar to DAN’s own Task 2 collapse despite DAN-DG never touching Sketch during training. The collapse is somewhat smaller for DAN-DG than for DAN, and that shared invariance corresponds to DAN-DG’s larger overall accuracy gain, but person collapsing into elephant and giraffe instead of spreading evenly argues that more invariance is not automatically better. Both methods also trained smoothly, unlike Task 2’s adversarial pair.

Task 3: ERM, DAN-DG, and SAM common comparison

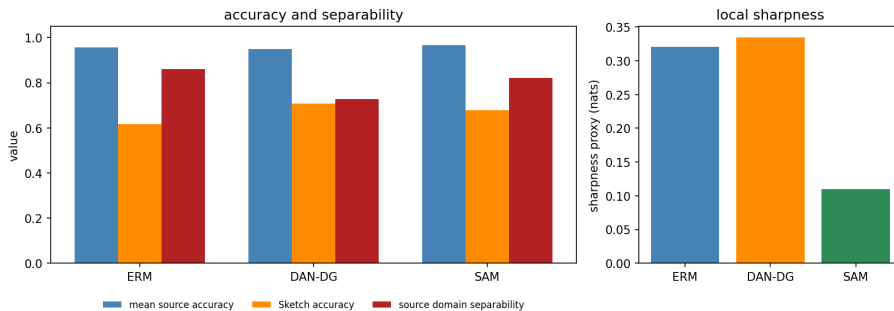


Figure 2: **DAN-DG and SAM both beat ERM on Sketch, but through entirely different mechanisms.** Mean source accuracy, Sketch accuracy, source separability, and the local sharpness proxy for ERM, DAN-DG, and SAM. SAM’s Sketch accuracy gain comes with a sharpness reduction of nearly a factor of three, while DAN-DG’s larger gain comes with no sharpness improvement and a visibly lower separability instead.

Class	DAN (Task 2, target aware)	DAN-DG (Task 3, target free)
dog	+0.034	+0.012
elephant	+0.161	+0.381
giraffe	+0.171	+0.331
guitar	-0.135	-0.109
horse	+0.038	-0.023
house	+0.138	+0.025
person	-0.819	-0.631

Table 5: Per class Sketch accuracy change relative to the shared ERM baseline, for Task 2’s target aware DAN and Task 3’s target free DAN-DG. Both share the same large gains on elephant and giraffe and the same person class collapse, despite only DAN having any access to Sketch during training.

Sweeping SAM’s perturbation radius ρ over 0.01, 0.05, and 0.10 gives mean source accuracy 0.959, 0.970, and 0.955, sharpness 0.234, 0.147, and 0.112, and Sketch accuracy 0.697, 0.748, and 0.715: sharpness falls monotonically as ρ rises, but Sketch accuracy peaks at the middle value tested. $\rho = 0.1$ ’s own training curve (Appendix Figure 6) shows why: mean validation macro-F1 sits at or below chance for four epochs before recovering, an instability that costs the accuracy gain the largest perturbation might otherwise have earned.

Sketch accuracy alone reveals DAN-DG’s alignment nearly erasing person; source validation never shows it. SAM cuts sharpness sharply, yet that ranking does not carry over to Sketch accuracy, since DAN-DG beats SAM despite no sharpness edge: each method’s own diagnostic supports only its own mechanism, and neither explains the other’s gain. Against Task 2’s target aware DAN, DAN-DG’s target free alignment still generalized better, though both share the same person failure regardless of target access.

6 Task 4: Open Set Recognition

6.1 Methodology

A CIFAR appropriate ResNet-18 is trained from random initialization on all ten CIFAR-10 classes (Vanilla), and again with RandAugment added (GCSC), to ask whether stronger positive augmentation improves unknown rejection alongside closed set accuracy. Four post hoc scores are then compared on the frozen Vanilla model, each one turning its ordinary closed set logits into a single unknownness score after training has already finished: maximum softmax probability [8], maximum logit score [19], energy [13], and Mahalanobis distance to the nearest class conditional Gaussian [11]. PROSER [20] takes the opposite approach, changing what the network is trained to do. It fine tunes the Vanilla checkpoint with five dummy classifiers under two objectives: a classifier placeholder loss that pushes one dummy to be the second most confident response once the true class is excluded, and a data placeholder loss that trains manifold mixup interpolations between two known classes’ features toward those dummies. Together these let the network learn unknown like regions without ever seeing real unknown data. Unknowns for every method come from two fixed CIFAR-100 groups, near and far, that never influence training, checkpoints, or thresholds; a threshold is calibrated once, on validation data, to accept 95 percent of known images.

6.2 Results

Mahalanobis distance is the best score overall, and on far unknowns specifically. MSP is a close second, and MLS and Energy come next, tracking the same raw logits within a point of each other. Mahalanobis’s edge is concentrated on far unknowns, suggesting feature space distance captures something raw logits do not, while near unknowns stay hard for every score alike precisely because they already resemble a known class. At the calibrated threshold, near unknown acceptance stays high for every score while far unknown acceptance is much lower.

GCSC’s RandAugment approach improves closed set accuracy over Vanilla and, under the same score, improves rejection on every axis too, most on near unknowns, the harder group, plausibly because stronger augmentation tightens each class’s feature region where similarity would otherwise let an unknown through. PROSER moves the opposite way: known class accuracy sits slightly below Vanilla’s, and MLS rejection is worse on every axis. Its validation accuracy also oscillated across its full run instead of converging smoothly the way Vanilla and GCSC’s did (Appendix Figure 7). This is evidence that manifold mixup interpolations are an imperfect stand in for real unknown data, though the placeholder idea itself may still hold up with a better interpolation scheme.

Score	Near AUROC	Far AUROC	All unknown AUROC
MSP	0.819	0.902	0.861
MLS	0.796	0.905	0.851
Energy	0.796	0.906	0.851
Mahalanobis	0.807	0.920	0.864

Model	Score	CSA	Near AUROC	Far AUROC	All unknown AUROC
Vanilla	MLS	0.947	0.796	0.905	0.851
GCSC	MLS	0.951	0.816	0.910	0.863
PROSER	MLS	0.942	0.789	0.876	0.832
PROSER	Placeholder	0.942	0.791	0.895	0.843

Table 6: Top: post hoc novelty scores on the frozen Vanilla model; every score separates far unknowns far more easily than near unknowns. Bottom: Vanilla, GCSC, and PROSER compared on closed set accuracy (CSA) and near/far/all unknown AUROC.

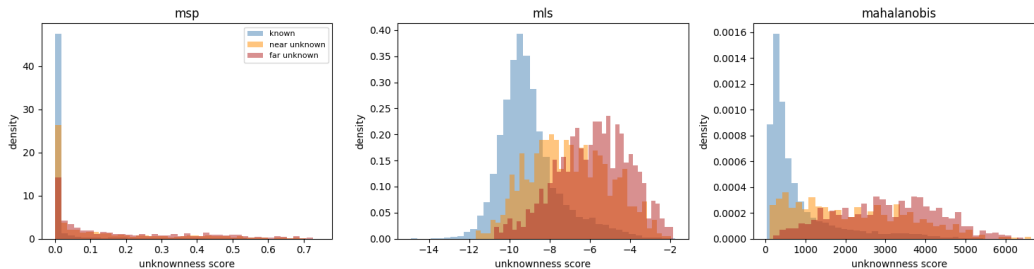


Figure 3: **Near unknown scores overlap heavily with known scores under every measure, while far unknown scores sit visibly apart.** Unknownness score distributions for known, near unknown, and far unknown images under MSP, MLS, and Mahalanobis.

The absorption pattern tracks semantic similarity closely. Among near unknowns, pickup_truck (94/100) and bus (82/100) are accepted almost as often as a known vehicle, absorbed mostly into automobile and truck, while motorcycle, least similar to any CIFAR-10 vehicle, is accepted least of the eight (41/100); wolf, fox, leopard, and camel absorb into cat, dog, deer, and horse. Far unknowns show the same mechanism more weakly: wardrobe’s boxy outline is often accepted as truck or cat (62/100), and a handful of the most confident far unknown failures have no obvious shared semantics at all, like a keyboard scored as truck or a chair scored as cat:

Group	Unknown class	Predicted known class	Score	Threshold
Near	pickup_truck	automobile	−11.67	−5.89
Near	pickup_truck	automobile	−11.50	−5.89
Near	bus	truck	−11.45	−5.89
Far	chair	cat	−10.75	−5.89
Far	keyboard	truck	−10.69	−5.89
Far	bowl	cat	−10.66	−5.89

Table 7: The most confidently accepted near and far unknown failures under the maximum logit score, with the vanilla model’s calibrated threshold for reference.

That absence of shared semantics suggests the decision boundary follows incidental texture and color correlations more than overall object structure. GCSC’s tightening of the feature space helps mainly on the near unknown group, since far unknowns were already easy to reject regardless of score or method; GCSC’s plain augmentation remains the more effective of the two closed set changes tried here for improving rejection.

7 Cross Task Discussion

Visual cues and distribution shift. Task 1 finds patch shuffling and cue conflict hardest for every backbone, both destroying global organization while leaving local structure intact. PACS’s Sketch domain removes an overlapping kind of evidence: texture and color are gone, leaving something closer to a pure outline. Guitar and house transfer best, because their outline alone nearly suffices; person, giraffe, and dog transfer worst, likely because they lean on the color and texture cues Task 1

showed these architectures partly depend on. This resemblance is only suggestive. The datasets and architectures involved differ too much to support a causal claim, but the pattern still points toward a single shared vulnerability: ordinary supervised or contrastive training leans on cues a real domain shift can remove entirely.

Invariance, discriminability, and target domain access. DAN and DAN-DG both make their domains harder to distinguish and both improve mean target accuracy, but both do so at the cost of nearly erasing person (Table 5). An overall feature distribution penalty has no way to tell domain separating information apart from class separating information, so this tradeoff is not surprising. SAM tells a contrasting story: a real gain with separability barely moved shows invariance is not the only route to generalization. Each method’s own diagnostic speaks only to its own mechanism, and no single diagnostic here explains every method’s gain. DAN-DG’s target free alignment also outperformed DAN’s target aware alignment, the opposite of what valuable target access should produce. A single benchmark cannot settle the general question, but target access is clearly no substitute for a well behaved objective.

Recognition, rejection, and the overall picture. Closed set accuracy and unknown rejection moved together for GCSC but apart for PROSER, echoing Tasks 1 and 2’s lesson that an aggregate number can hide the evidence that explains behavior. The near unknown group was chosen for semantic closeness to known classes, and it is consistently the harder case across every score and method. This is exactly the case in which a confident classifier should sometimes decline to answer.

Limitations. One limitation is that results depend on a single training run rather than an average over multiple runs. Task 3 showed this matters: SAM’s own ρ sweep at $\rho = 0.1$ goes below chance for several epochs before recovering (Appendix Figure 6), an instability large enough that a small gap between two methods, such as DAN-DG’s edge over SAM on Sketch, could just as easily come from random training variation as from a real difference between the methods. A second limitation is scope: each paradigm here depends on one benchmark and one backbone family, so the specific rankings between methods, DAN-DG over SAM, or Mahalanobis over MSP, should be read as evidence for this setting; extending them into a general claim would need testing across multiple benchmarks.

8 Conclusion

Clean accuracy alone is not enough to trust a vision model, and these four experiments show why in different ways. ResNet, ViT, and CLIP reach near identical clean accuracy while relying on visibly different evidence once color, shape, texture, and layout are manipulated. DAN-DG and SAM reach a similar Sketch accuracy gain over the shared ERM baseline for entirely different reasons, one by lowering feature separability across source domains and the other by flattening the loss surface, and neither method’s own diagnostic explains the other’s success. DAN-DG even outperforms Task 2’s target aware DAN on Sketch despite never touching the target domain during training, so target access is no substitute for a well behaved objective. Closed set accuracy and unknown rejection moved together for GCSC but apart for PROSER, so a classifier’s performance on its own training distribution says nothing about how well it rejects what falls outside it. Building a robust vision system is less about maximizing a single number and more about choosing what to preserve, what to treat as invariant, and when to refuse to answer.

References

- [1] Adam Coates, Andrew Ng, and Honglak Lee. An analysis of single-layer networks in unsupervised feature learning. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2011.
- [2] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.
- [3] Pierre Foret, Ariel Kleiner, Hossein Mobahi, and Behnam Neyshabur. Sharpness-aware minimization for efficiently improving generalization. In *International Conference on Learning Representations (ICLR)*, 2021.
- [4] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35, 2016.
- [5] Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A. Wichmann, and Wieland Brendel. Imagenet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. In *International Conference on Learning Representations (ICLR)*, 2019.
- [6] Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations (ICLR)*, 2021.
- [7] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [8] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations (ICLR)*, 2017.
- [9] Xun Huang and Serge Belongie. Arbitrary style transfer in real-time with adaptive instance normalization. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.
- [10] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- [11] Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [12] Da Li, Yongxin Yang, Yi-Zhe Song, and Timothy M. Hospedales. Deeper, broader and artier domain generalization. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.
- [13] Weitang Liu, Xiaoyun Wang, John D. Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [14] Mingsheng Long, Yue Cao, Jianmin Wang, and Michael I. Jordan. Learning transferable features with deep adaptation networks. In *International Conference on Machine Learning (ICML)*, 2015.
- [15] Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I. Jordan. Conditional adversarial domain adaptation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [16] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021.
- [17] Maithra Raghu, Thomas Unterthiner, Simon Kornblith, Chiyuan Zhang, and Alexey Dosovitskiy. Do vision transformers see like convolutional neural networks? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- [18] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9:2579–2605, 2008.

- [19] Sagar Vaze, Kai Han, Andrea Vedaldi, and Andrew Zisserman. Open-set recognition: a good closed-set classifier is all you need? In *International Conference on Learning Representations (ICLR)*, 2022.
- [20] Da-Wei Zhou, Han-Jia Ye, and De-Chuan Zhan. Learning placeholders for open-set recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

A Appendix

DAN and CDAN’s confusion matrices (Section 4) are shown here in full, along with training curves for SAM at $\rho = 0.1$ and PROSER (Sections 5 and 6), both of which show unstable training.

A.1 DAN

Figure 4 shows DAN’s confusion matrix on Sketch as a row percentage. Person sketches are predicted house 85 percent of the time, and dog and elephant each lose roughly a third of their images to person the same way; every other class stays mostly on its own diagonal.

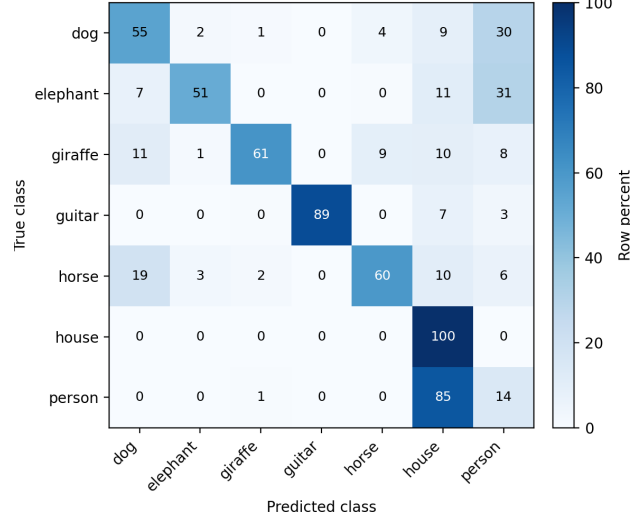


Figure 4: DAN’s confusion matrix on Sketch, as a percentage of each true class’s images.

A.2 CDAN

Figure 5 shows CDAN’s confusion matrix on Sketch as a row percentage. Every class outside dog gets predicted dog most of the time except house, which splits 40 percent dog and 60 percent correctly as house; elephant and person are predicted dog for every single image.

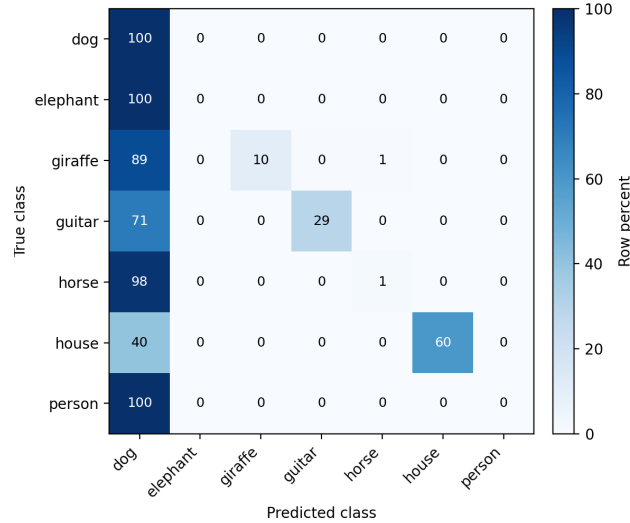


Figure 5: CDAN’s confusion matrix on Sketch, as a percentage of each true class’s images.

A.3 SAM at $\rho = 0.1$

Figure 6 shows classification loss and mean source validation macro F1 for the largest perturbation radius swept in Section 5. Validation macro F1 sits at or below chance for the first four epochs

before recovering, an instability that costs this setting the accuracy edge a larger perturbation might otherwise earn.

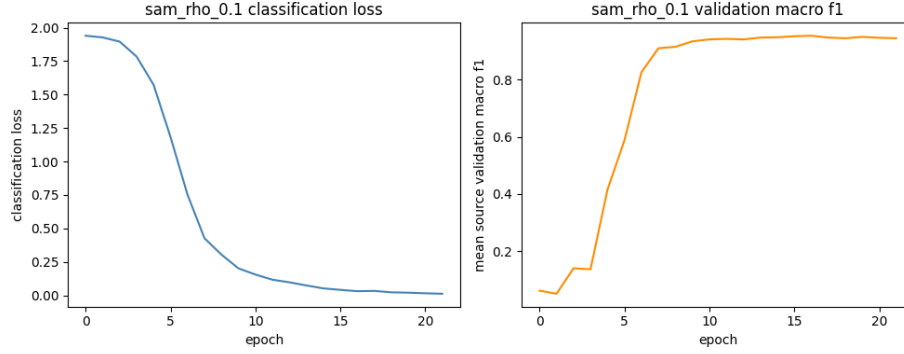


Figure 6: SAM’s training curves at $\rho = 0.1$.

A.4 PROSER

Figure 7 shows training loss, validation accuracy, and data placeholder loss across PROSER’s full fifty epoch fine tuning run. Validation accuracy oscillates for the entire run instead of converging smoothly the way Vanilla and GCSC’s own training curves do.



Figure 7: PROSER’s training curves across its full fine tuning run.